BOOSTING MACHINE LEARNING PERFORMANCE WITH INTEL® DATA ANALYTICS ACCELERATION LIBRARY (INTEL® DAAL)

Intel, contrary to what you might expect, was not named in remembrance of the word 'intelligence', but rather, as a portmanteau of 'integrated' and 'electronics', a homage to the Integrated Circuits whose development ushered in the age of electronics. One might be especially easily forgiven for this mistake today, when Intel is taking on the mantle of a frontrunner in artificial intelligence, the foundational tools and methods of which are in machine learning.

In 2015, Intel launched its Data Analytics Acceleration Library, popularly known as Intel® DAAL (as one might address one's darling, and as a data scientist, as one certainly should). As far as names go, there's no catfishing here. Intel® DAAL offers exactly what it says on the tin - high-performance data analysis algorithms implemented in significantly more time- and space-efficient ways, with the added benefit of the word 'Library' not being restricted to any single platform or programming language. Mac vs Linux, or God forbid, Windows? Intel® DAAL works for you. R-lover? Pythonista? MATLAB enthusiast? Intel® DAAL works for you.

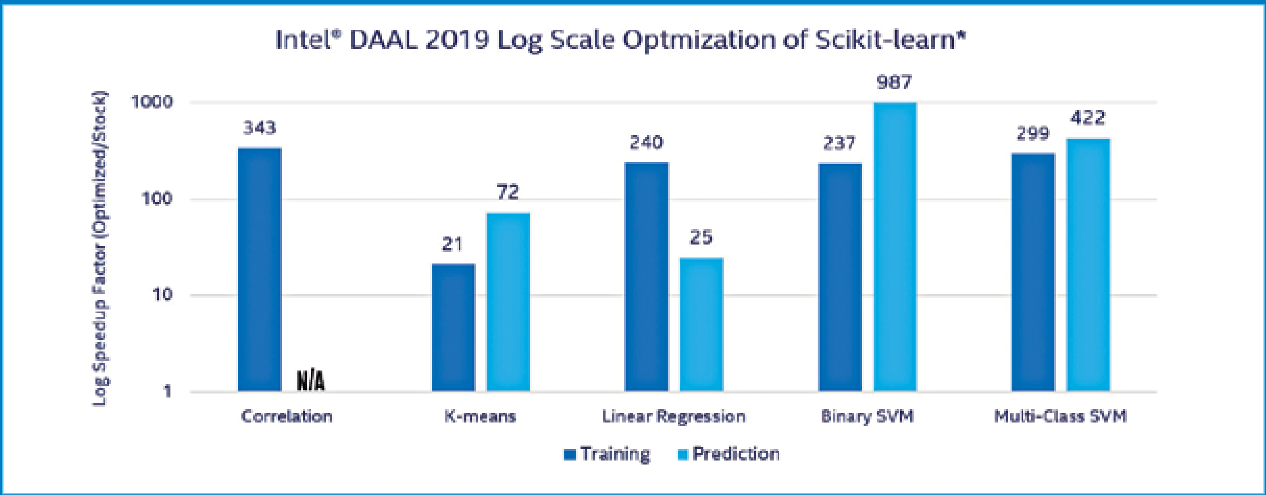
Intel® DAAL has entered a market that's already relatively saturated, both by competition, as well as by steep learning curves. It takes time to learn a new library. It takes longer to learn how to perform analysis on data. And it usually takes the longest of these to run the actual algorithms on a platform of one's comfort. With Intel® DAAL, Intel has shown an ability to read the market well, offering competitive advantages in each of these areas. Intel® DAAL is easy to integrate with popular programming languages. It offers extensive documentation and lots of Intel-supported Data Analytics learning resources. And the cherry on top is that it's significantly faster than most popular implementations of most other algorithms. The reason is simple - Intel has integrated its wealth of knowledge on hardware integration with its significant share of the processor market to figure out how to speed everything up by a factor anywhere between 2 and 1000.

WHY YOU SHOULD USE INTEL® DAAL

Intel® DAAL accelerates the performance of almost any ML algorithm and thereby improves accuracy. Training in ML is often an iterative process, and one usually truncates the process for some acceptable accuracy threshold due to CPU time constraints. For instance, many top firms rebuild an XGBoost model overnight to incorporate information from the previous day - this build has to be optimised not just for accuracy, but for computing resources. With computing speeds that orders of magnitude higher, Intel® DAAL-offers a huge competitive advantage for these kinds of applications. It isn't just speed - Intel® DAAL is space-optimized, and can analyse significantly larger datasets with similar computing power. Applications that truncate or compress their data (perhaps using PCA or other dimensionality reduction algorithms) can leverage Intel® DAAL to retain more of the fine-grained information in their training data. Every function in Intel® DAAL is specifically tuned to the instruction set, vector width, core count and memory structure of its intended processor.

As far as usage support goes, Intel® DAAL-supports offline, streaming, as well as distributed usage models. Whether your application makes use of cloud services, cluster computing, or even a simple local machine, rest assured that Intel® DAAL will integrate well with your system. This is another place where Intel's relative dominance on the processor market works well in your favour. Chances are you're using Intel hardware, and therefore, chances are that Intel® DAAL is exactly what you need.

Finally, Intel® DAAL optimises your analysis at every step - from data ingestion to algorithm computation - and does so in synergy, as the brilliant minds at Intel have designed it to do. Never again will you need



Intel® DAAL's daal4py offering even provides a Python API for its algorithms